Just Tell Them by Dr. Zach Groshell **Planning Summary by Helen Reynolds (@helenrey/bsky)**

Chapter	Do this/Learn this
Intro/ Why Explain: Explaining is enabling others to 'borrow' information from us and reorganize it, Students don't just absorb – they have a cognitive architecture (working memory (WM), long-term memory (LTM), WM can be overloaded (cognitive load (CL) theory) but is helped by LTM, direct instruction (DI) reduces CL, increases learning	
Explain with undivided attention: Set the conditions for focus None of what follows will work if students aren't listening, visual and auditory distractions take up space in WM, how they enter the room sets the scene – control it, if you do the seating it removes pressure on them to choose, anything more than silence introduces distraction to WM	
Explain clearly: Say *only* the words that need to be said **Avoid**: 1. Vagueness by trimming unnecessary words which clog up WM, avoid 2. Mazes (false starts/ hesitations/unnecessary repetitions), 3. Discontinuities (information that is irrelevant, or relevant but not now) 4. Seductive details (interesting or entertaining content that distracts)	
Explain interactively: Alternate between inputs and outputs 'Extended lecture' can be as frustrating as 'discovery learning', effective explaining = managing attention + behaviour + CL by chunking and alternating/ doing example/problem pairs, it's a dialogue – take turns, needs a brisk pace to increase number of responses, which informs pace, we want our students to do hard thinking which you can get from cold-calling, increasing think time and explaining to themselves (self-elaboration) and to others.	
Explain with visuals: Speaking with images and drawings Spoken word is transient – we need to increase processing time using images/diagrams/graphics, there are 2 simultaneous channels into working memory = dual coding via spoken word + visuals, many guiding principles, press B during PowerPoint to get a black screen then switch to whiteboard for dynamic drawing.	

Explain with examples: Show what it is and what it isn't 1. Examples are specific instances of material; explanations situate examples in wider context/ concepts, and need non-examples, and erroneous examples, choice of examples used is VERY important to exemplify boundaries of concept. 2. Deep understanding = memory from a long time ago or ability to apply in a wide range of contexts such as learning from errors. 3. Analogies should capitalize on prior knowledge, highlight shared structure, not overload WM, beware getting them to generate them	
Explain with stories: Narrative, emotion, gestures Engagement is NOT entertainment, but teaching is performance, 1. Storytelling is psychologically privileged, 2. Emotional design – emotionally resonant speech includes personalization and other features to engage on emotional level 3. Gestures are co-speech, more than just fun, good for communicating visuo-spatial information related to embodiment and social agency	
Explain and release): Gradually fade guidance What you give them will depend on their level of expertise, there is a novice ↔ expert continuum which is domain-specific so instructional methods need to change gradually while managing cognitive load, 1. Going from 'I do' to 'you do' 2. Going from 'out loud' to 'in their heads', 3. Going from simple to complex (managing element interactivity – the juggling in WM) 4. Going from massed practice to spaced practice.	
Conclusion: We may have embraced minimal guidance/ guide on the side ideas but explicit instruction works, particularly for "low" or "at-risk" students.	

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